Q) Create a structure variable student and estimate the size of the structure manually, then write a c program to find size of structure using sizeof operator. Now observe both the o/ps. Are they same or different, justify your statement.

returns to whom.

Q1. Ans: Structure definition

```
struct student
{
    int rollno;
    char grade;
    float marks;
};
```

Size estimation

int roll = 4 bytes
char grade = 1 byte
float marks = 4 bytes
size = 4+1+4 = 9 bytes

C programming using sizeof:

```
#include<stdio.h>
struct student
{
    int rollno;
    char grade;
    float marks;
};
int main()
{
    struct student s;
    Printf ("size = %d bytes\n", sizeof (s));
    return 0;
}
```

∴ O/P are different because compiler adds padding btw structure members so that data is stored at memory addresses that are efficient for processor to access. Although actual data occupies 9 bytes, compiler inserts 3 extra bytes, making it 12 bytes